



Computer Networks

Pushpendra Singh

The Course

- Computer Networks – Sec B (non-CSE)
 - Sec A is being taught by Prof. Rinku Shah
- The syllabus, assignments, and exams will be same.
- Teaching and other in-class policies would be different.
- For sec B, attendance is mandatory.
- Plagiarism Policy of the institute will be followed
 - First instance: letter grade reduction
 - Second instance/cheating in proctored exams:
F
- Instructor
 - Pushpendra Singh
 - Professor, CSE
 - A-502, R&D
 - A-515, Mélange Lab
 - Office hours for the course
 - Mon, Wed: 4:30-5:00 p.m.

The Course

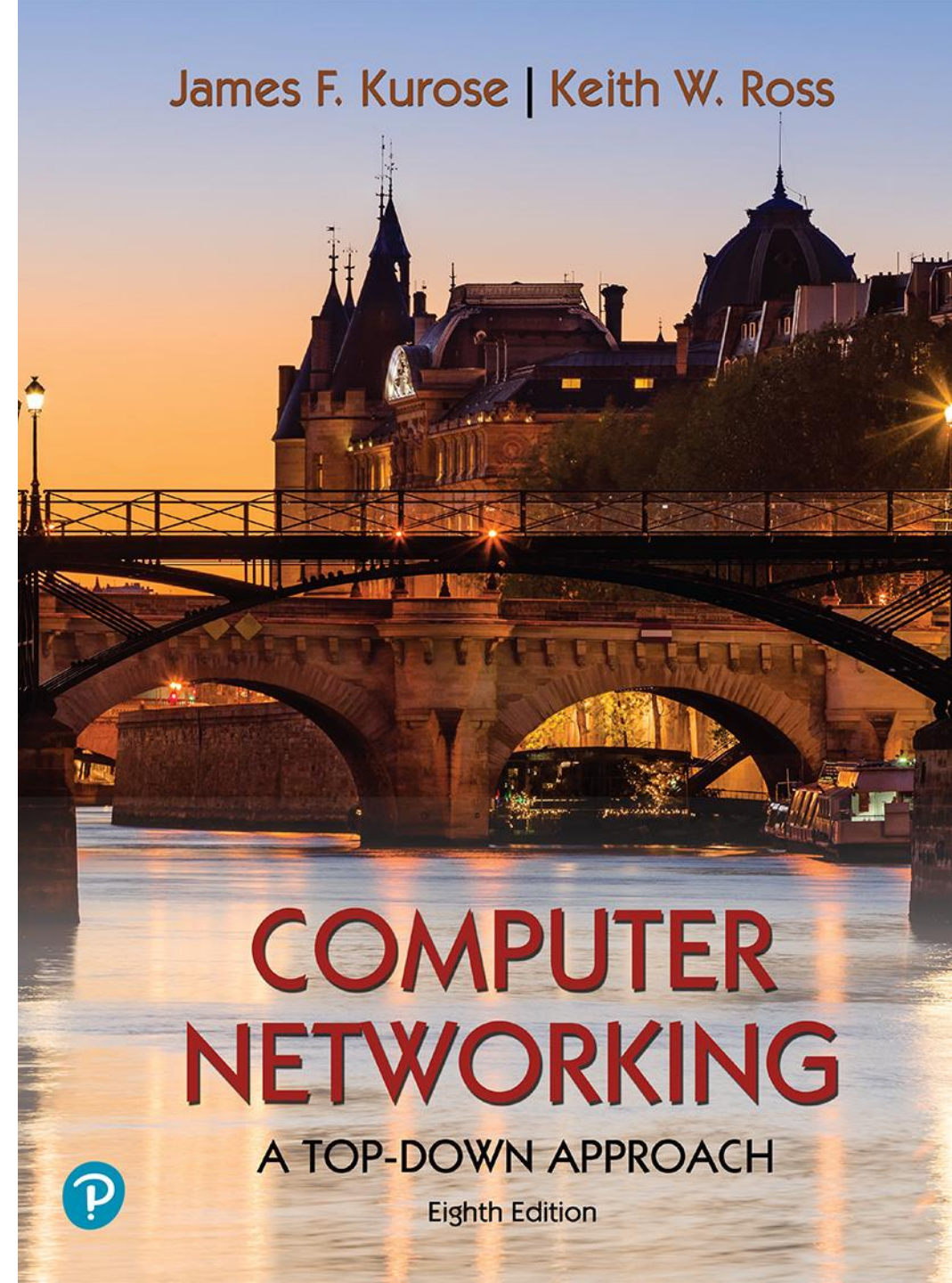
Grading Component	Tentative Count	Weightage
Class quiz	Best 2 out of 3	10%
Assignments	4	40%
Mid-sem	1	15%
End-sem	1	35%

No extension of Assignment Dates once announced

ZERO marks for late submission – Google classroom classifying it as late

The Course

- Textbook
 - Kurose & Ross, “Computer Networking: A top-down approach, 8th edition”
 - We will also use slides from Kurose and Ross
 - Slides may be modified



Chapter 1: introduction

Chapter goal:

- Get “feel,” “big picture,” introduction to terminology
 - more depth, detail *later* in course



Overview/roadmap:

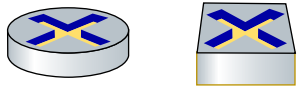
- What *is* the Internet? What *is* a protocol?
- **Network edge:** hosts, access network, physical media
- **Network core:** packet/circuit switching, internet structure
- **Performance:** loss, delay, throughput
- Protocol layers, service models
- Security
- History

The Internet: a “nuts and bolts” view



Billions of connected computing *devices*:

- *hosts* = end systems
- running *network apps* at Internet's “edge”



Packet switches: forward packets (chunks of data)

- *routers, switches*

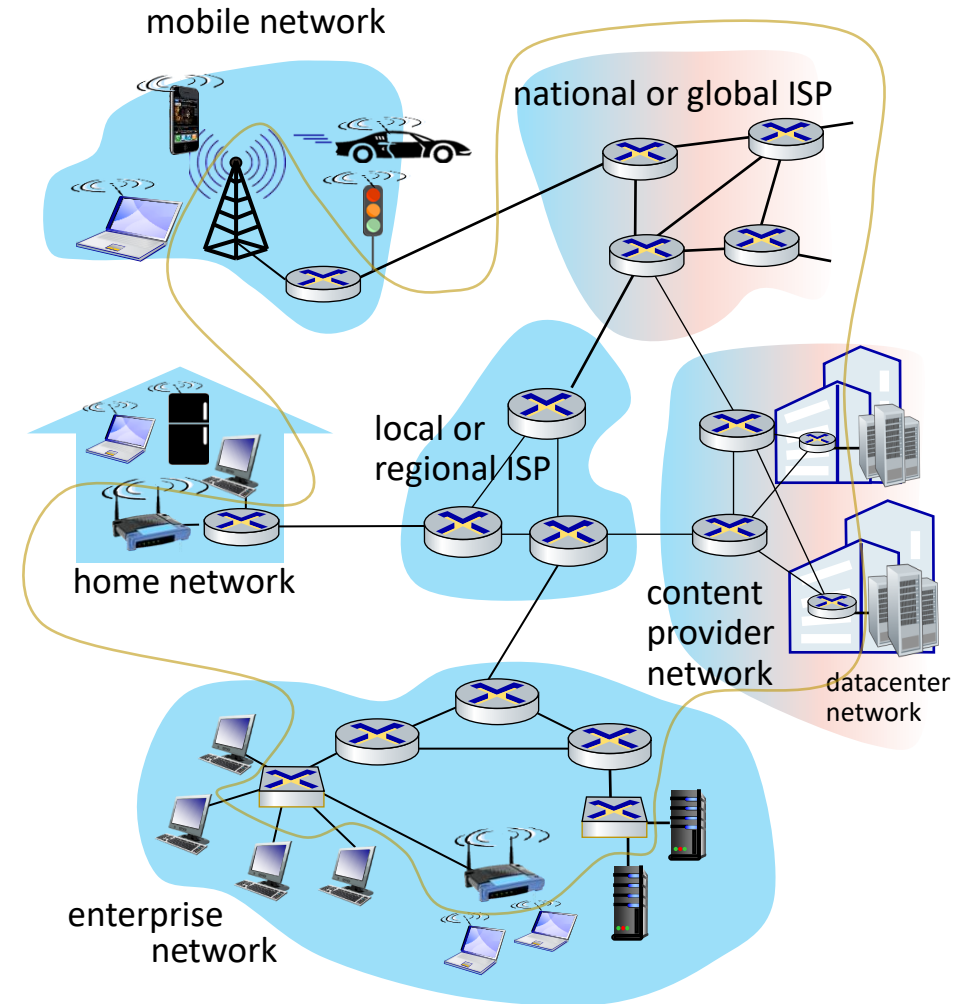


Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

Networks

- collection of devices, routers, links: managed by an organization



“Fun” Internet-connected devices



Amazon Echo



Internet refrigerator



Security Camera



Internet phones



Gaming devices



IP picture frame



Slingbox: remote control cable TV



Pacemaker & Monitor



Web-enabled toaster + weather forecaster



sensorized, bed mattress



AR devices

Fitbit



diapers



Tweet-a-watt:
monitor energy use



bikes



cars



scooters

Others?

The Internet: a “nuts and bolts” view

- *Internet: “network of networks”*

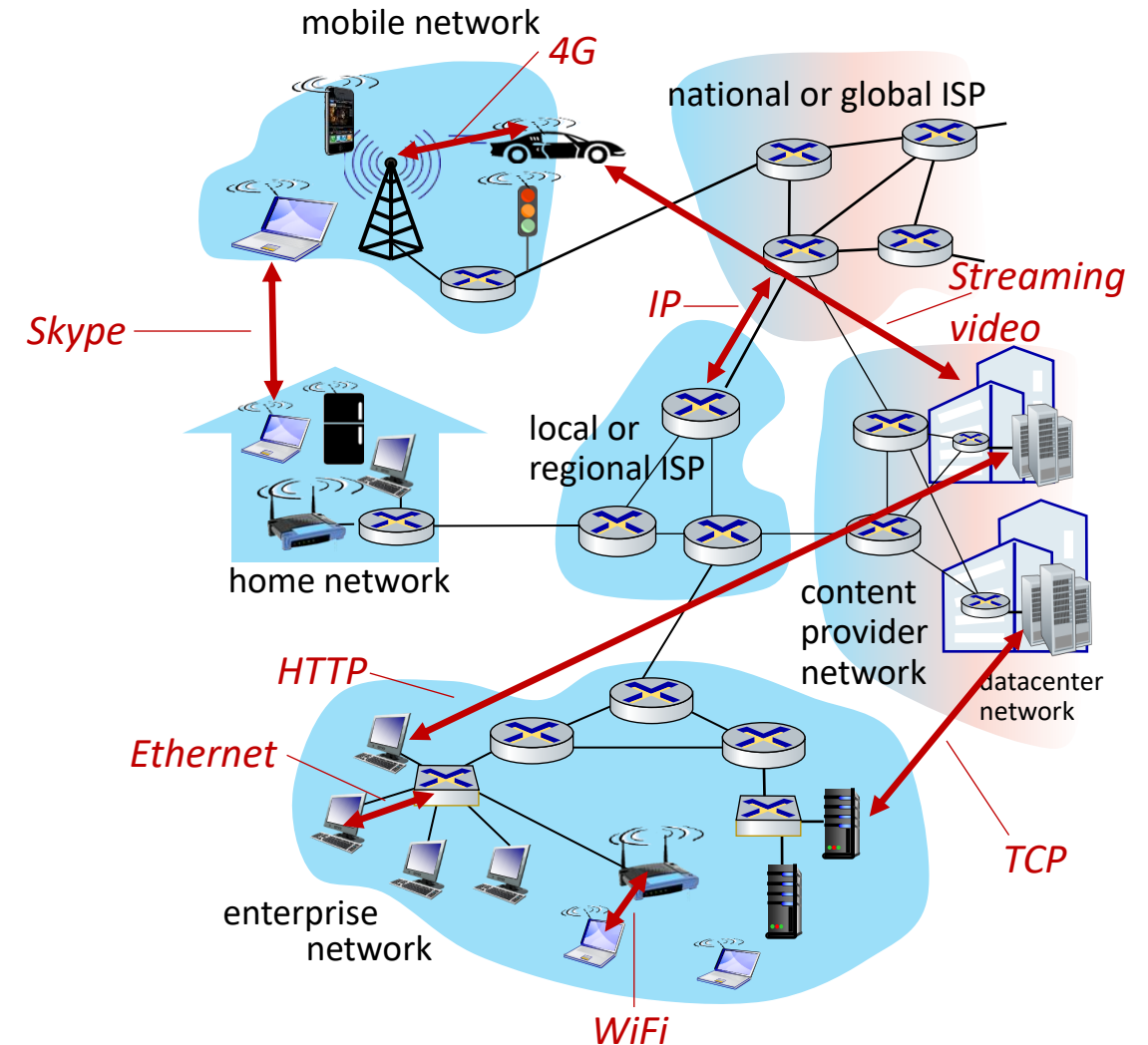
- Interconnected ISPs

- *protocols are everywhere*

- control sending, receiving of messages
- e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4/5G, Ethernet

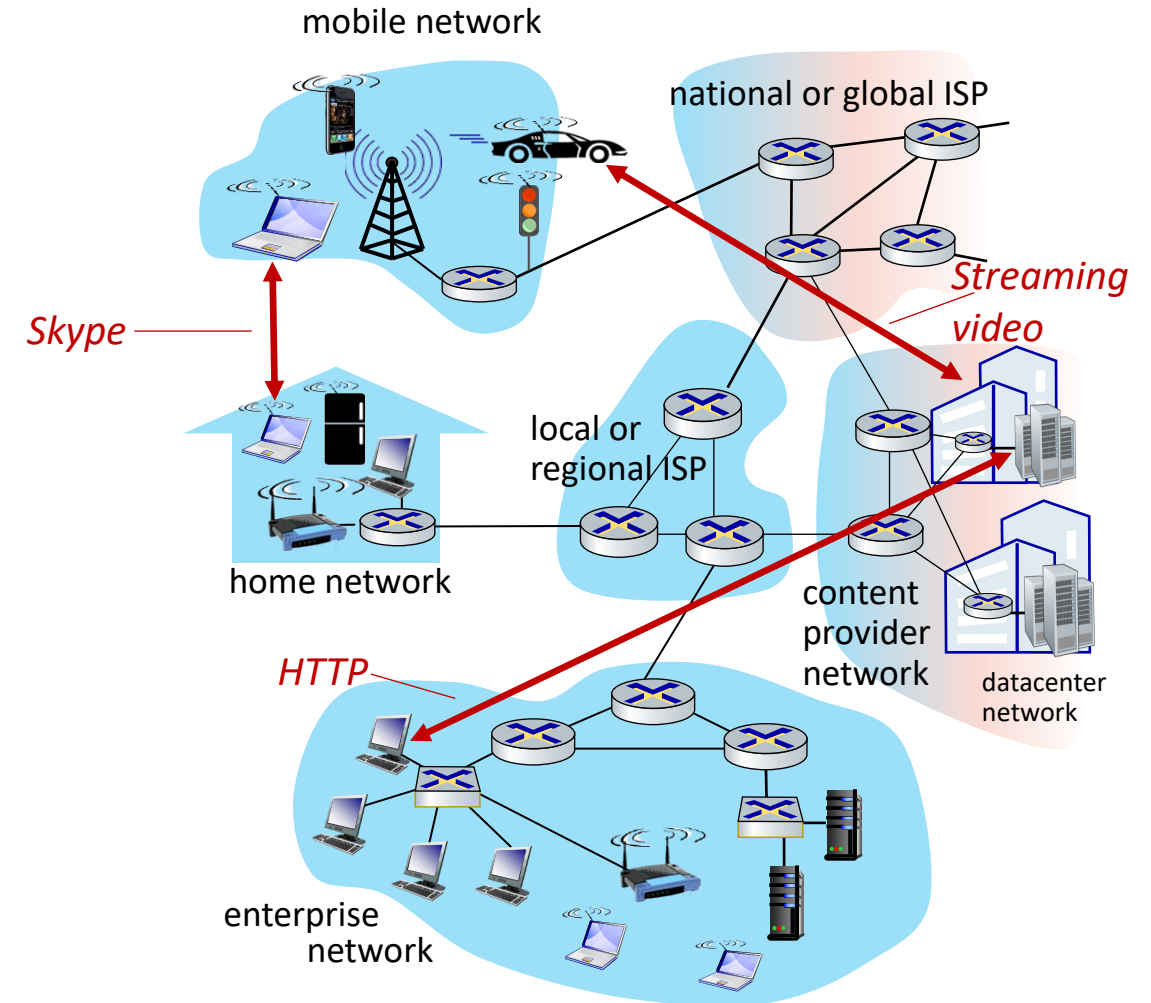
- *Internet standards*

- RFC: Request for Comments
- IETF: Internet Engineering Task Force



The Internet: a “services” view

- *Infrastructure* that provides services to applications:
 - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...
- provides *programming interface* to distributed applications:
 - “hooks” allowing sending/receiving apps to “connect” to, use Internet transport service
 - provides service options, analogous to postal service



What's a protocol?

Human protocols:

- “what’s the time?”
- “I have a question”
- introductions

Rules for:

- ... specific messages sent
- ... specific actions taken
when message received,
or other events

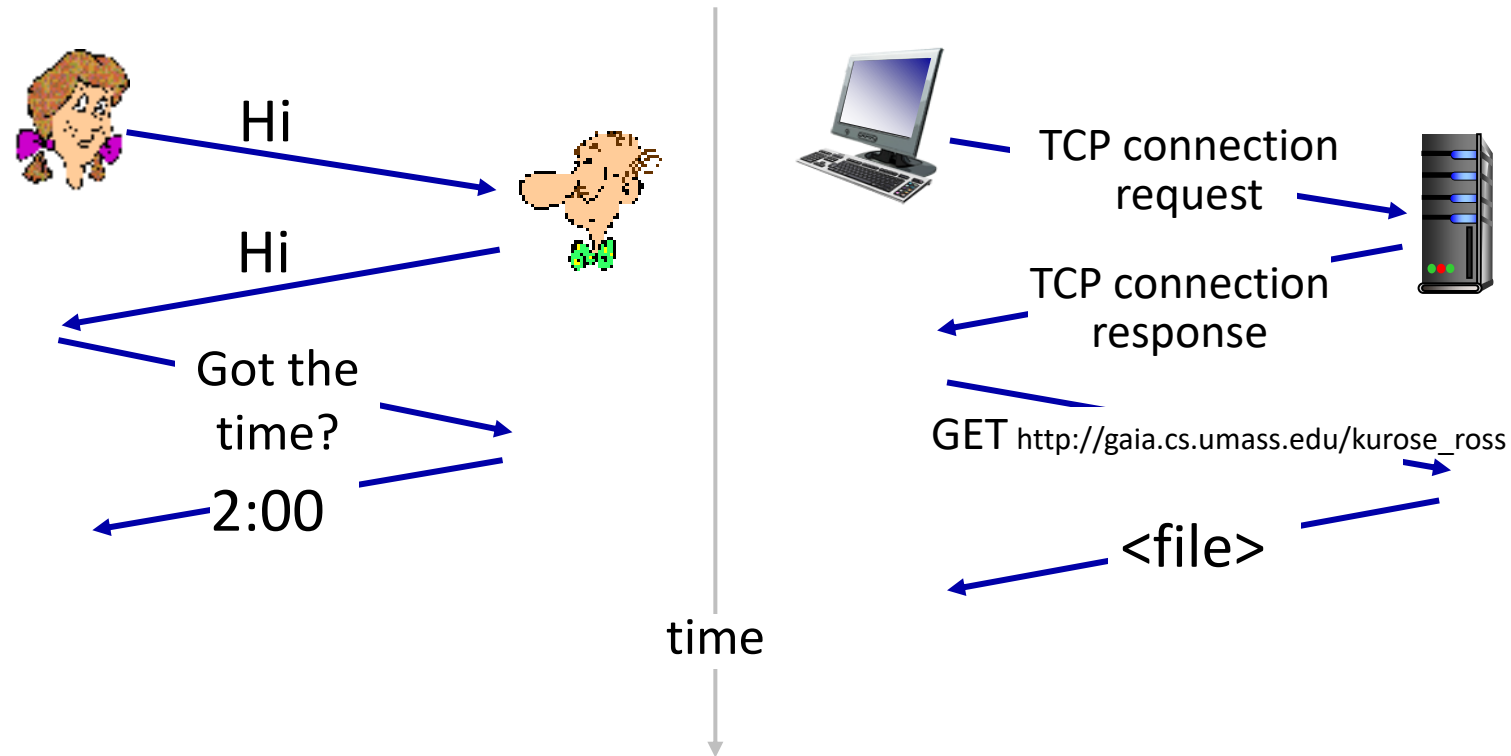
Network protocols:

- computers (devices) rather than humans
- all communication activity in Internet
governed by protocols

*Protocols define the **format, order** of
messages sent and received among
network entities, and **actions taken**
on message transmission, receipt*

What's a protocol?

A human protocol and a computer network protocol:



Q: other human protocols?

Chapter 1: roadmap

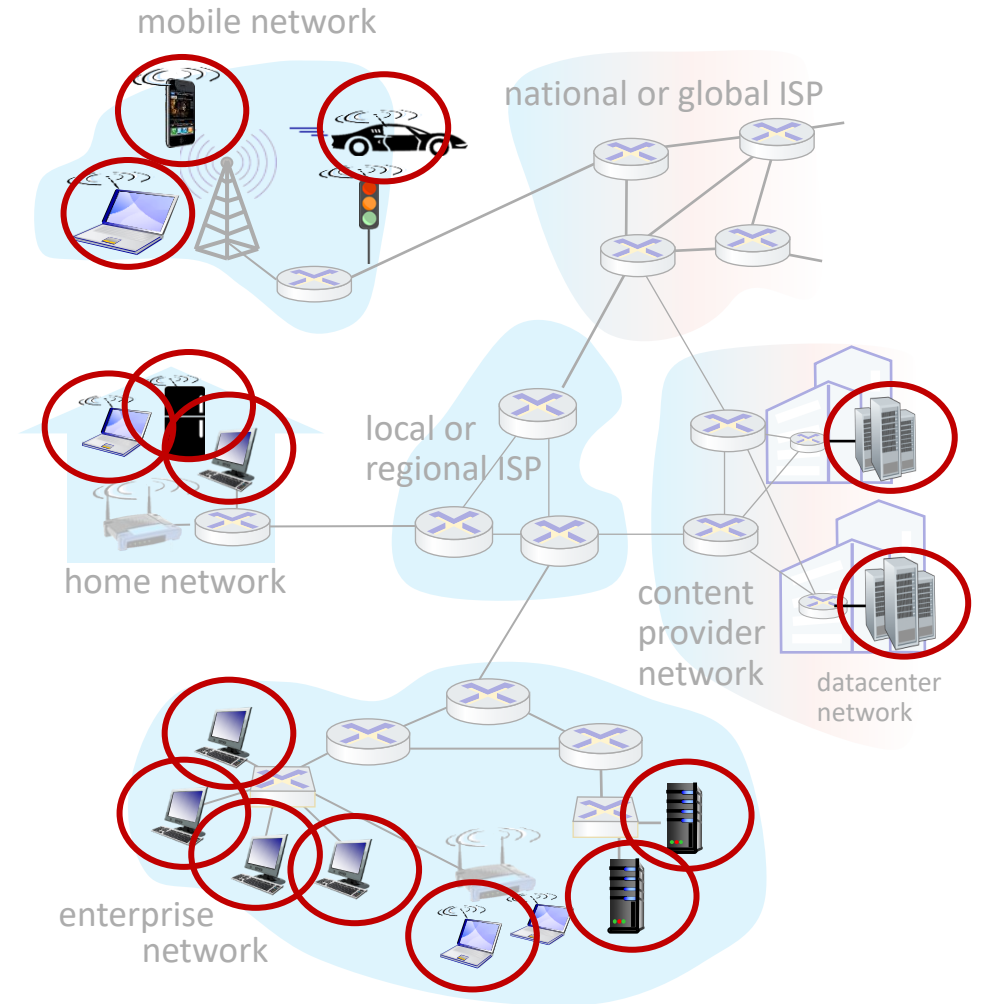
- What *is* the Internet?
- What *is* a protocol?
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A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers



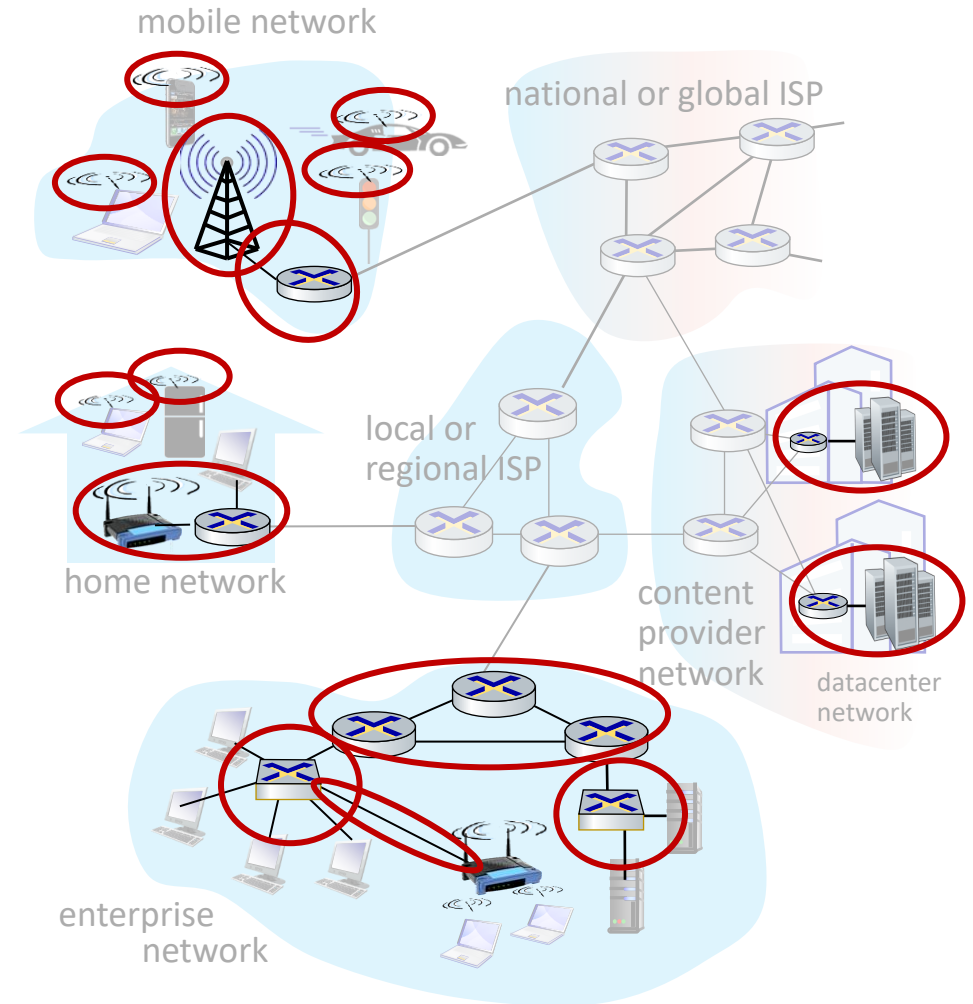
A closer look at Internet structure

Network edge:

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Access networks, physical media:

- wired, wireless communication links



A closer look at Internet structure

Network edge:

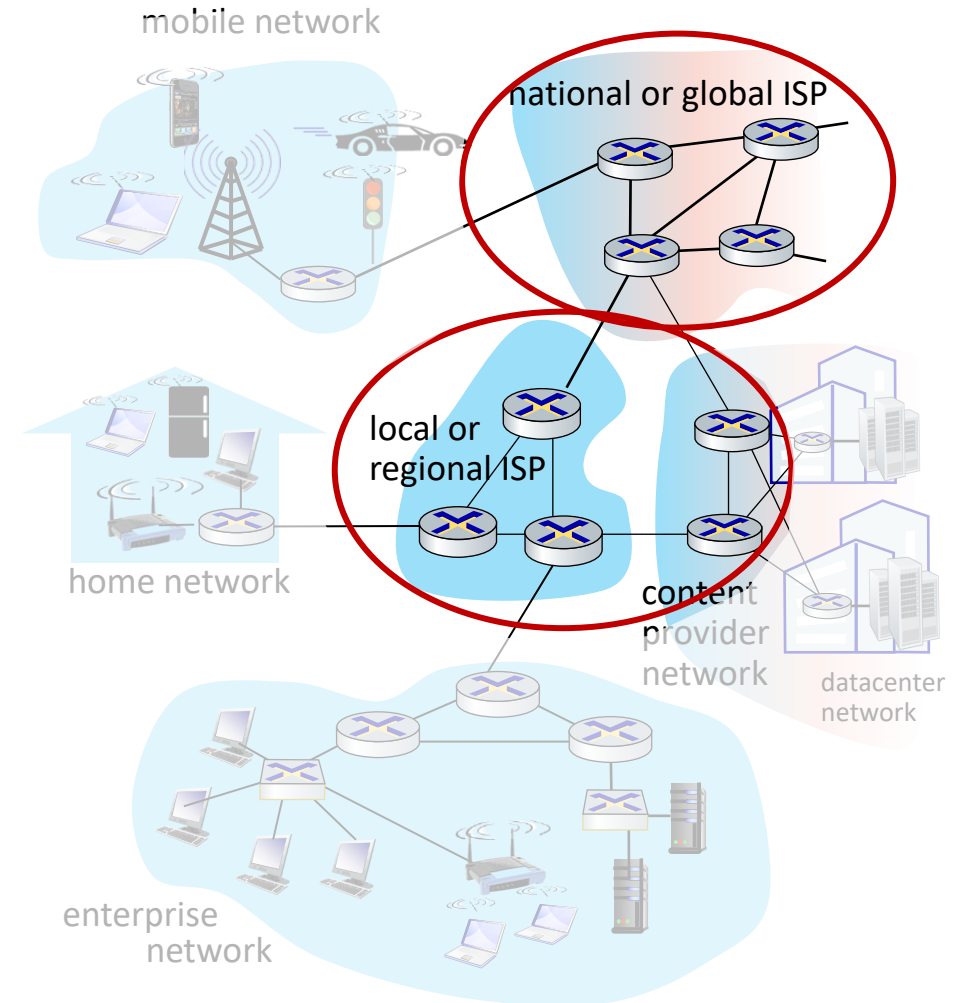
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Access networks, physical media:

- wired, wireless communication links

Network core:

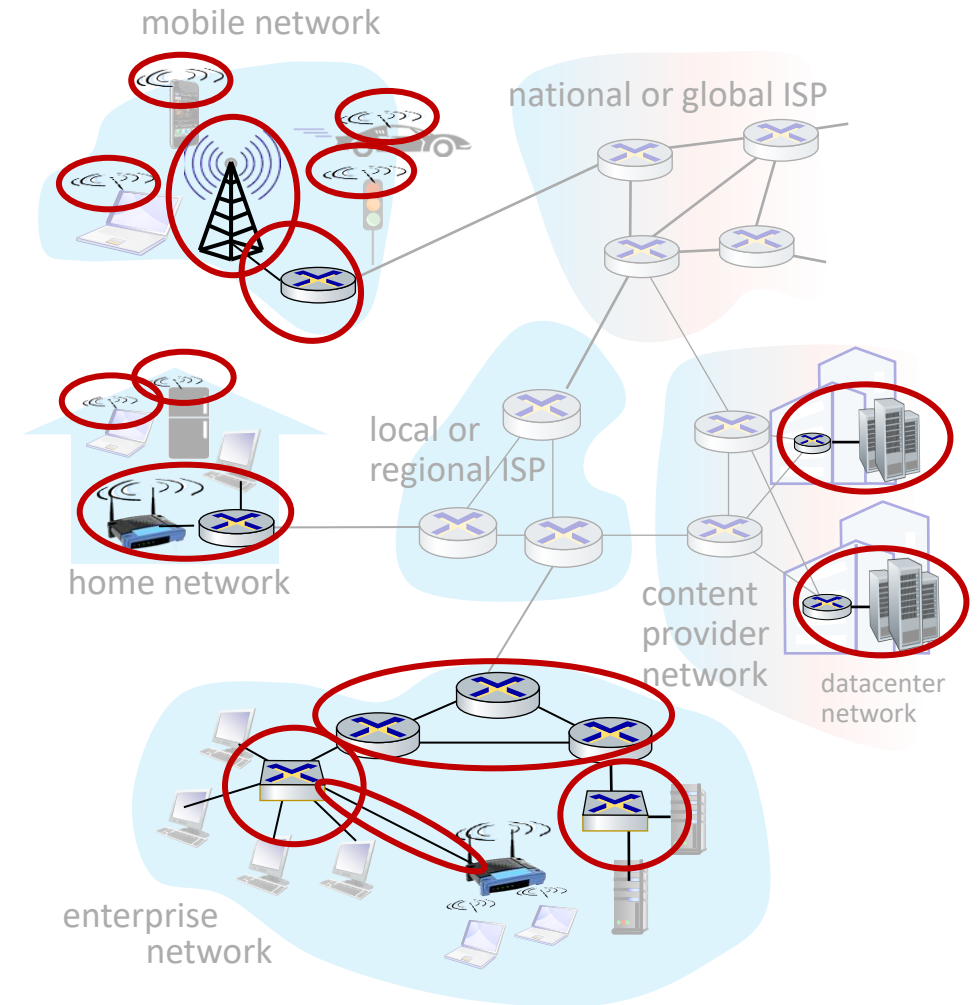
- interconnected routers
- network of networks



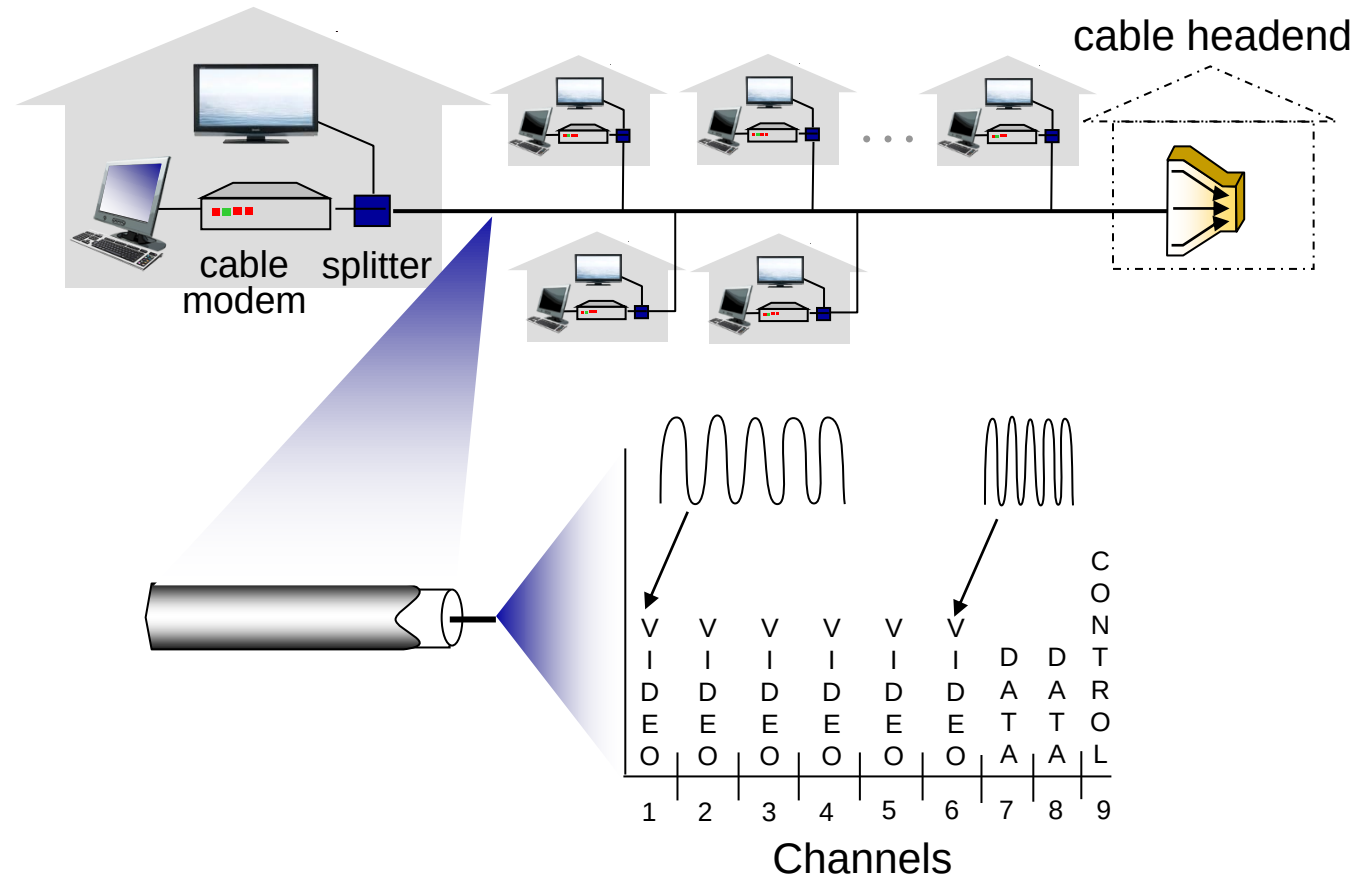
Access networks and physical media

Q: How to connect end systems to edge router?

- residential access nets
- institutional access networks (school, company)
- mobile access networks (WiFi, 4G/5G)

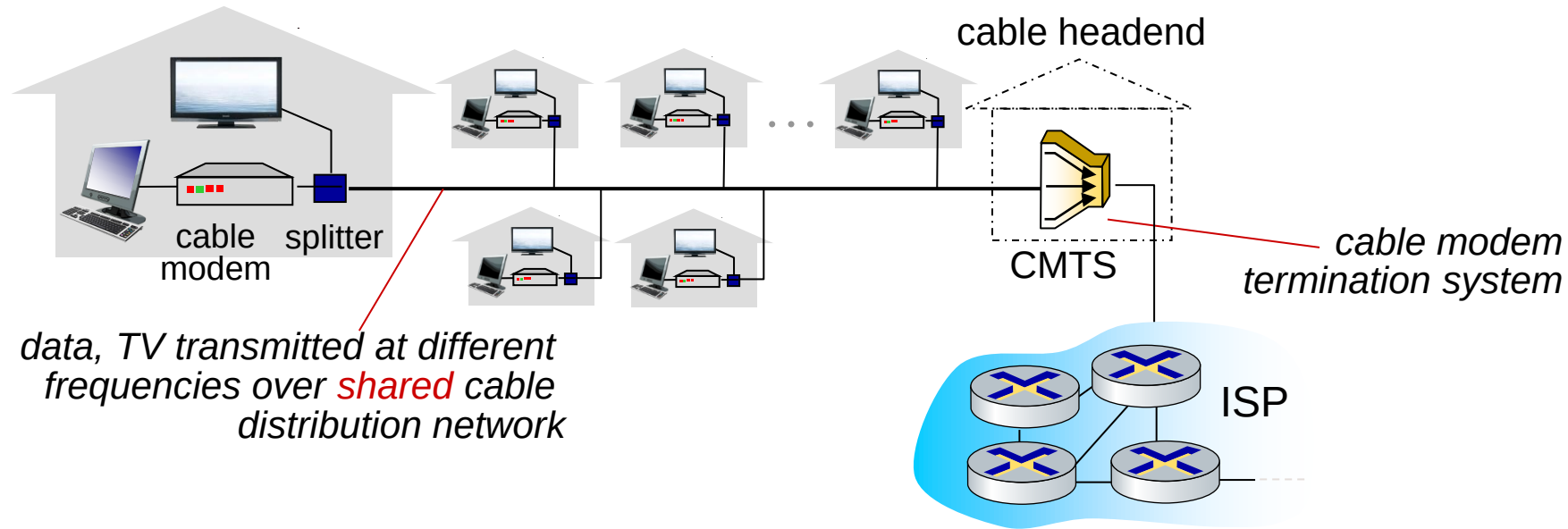


Access networks: cable-based access



frequency division multiplexing (FDM): different channels transmitted in different frequency bands

Access networks: cable-based access



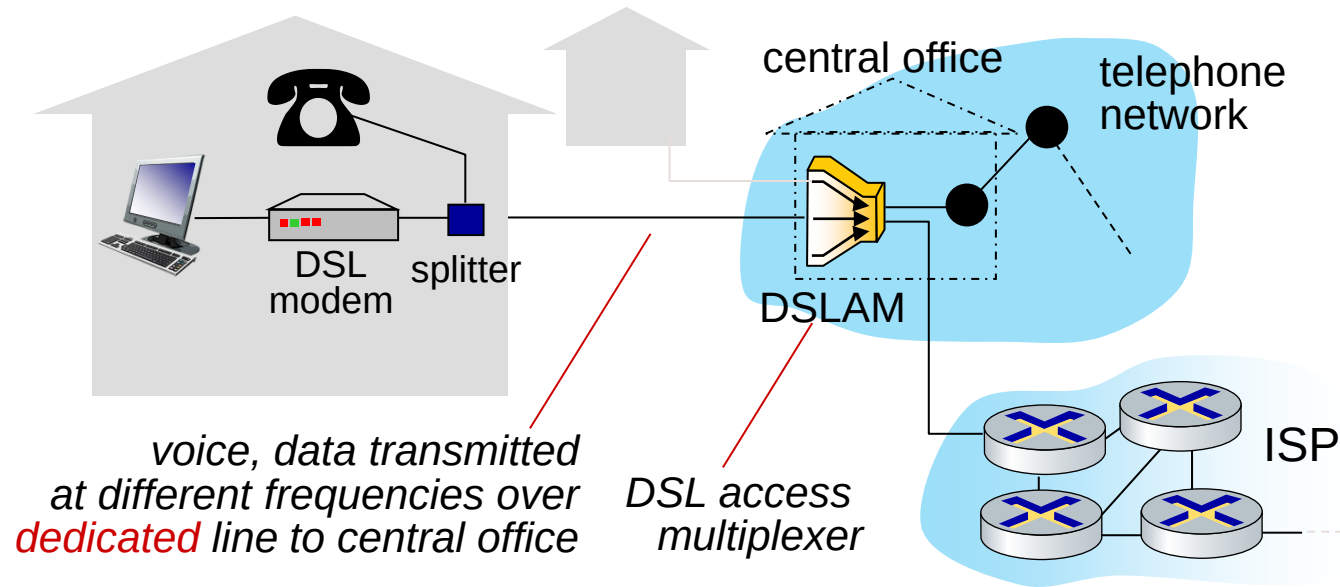
■ HFC: hybrid fiber coax

- asymmetric: up to 40 Mbps – 1.2 Gbps downstream transmission rate, 30-100 Mbps upstream transmission rate

■ network of cable, fiber attaches homes to ISP router

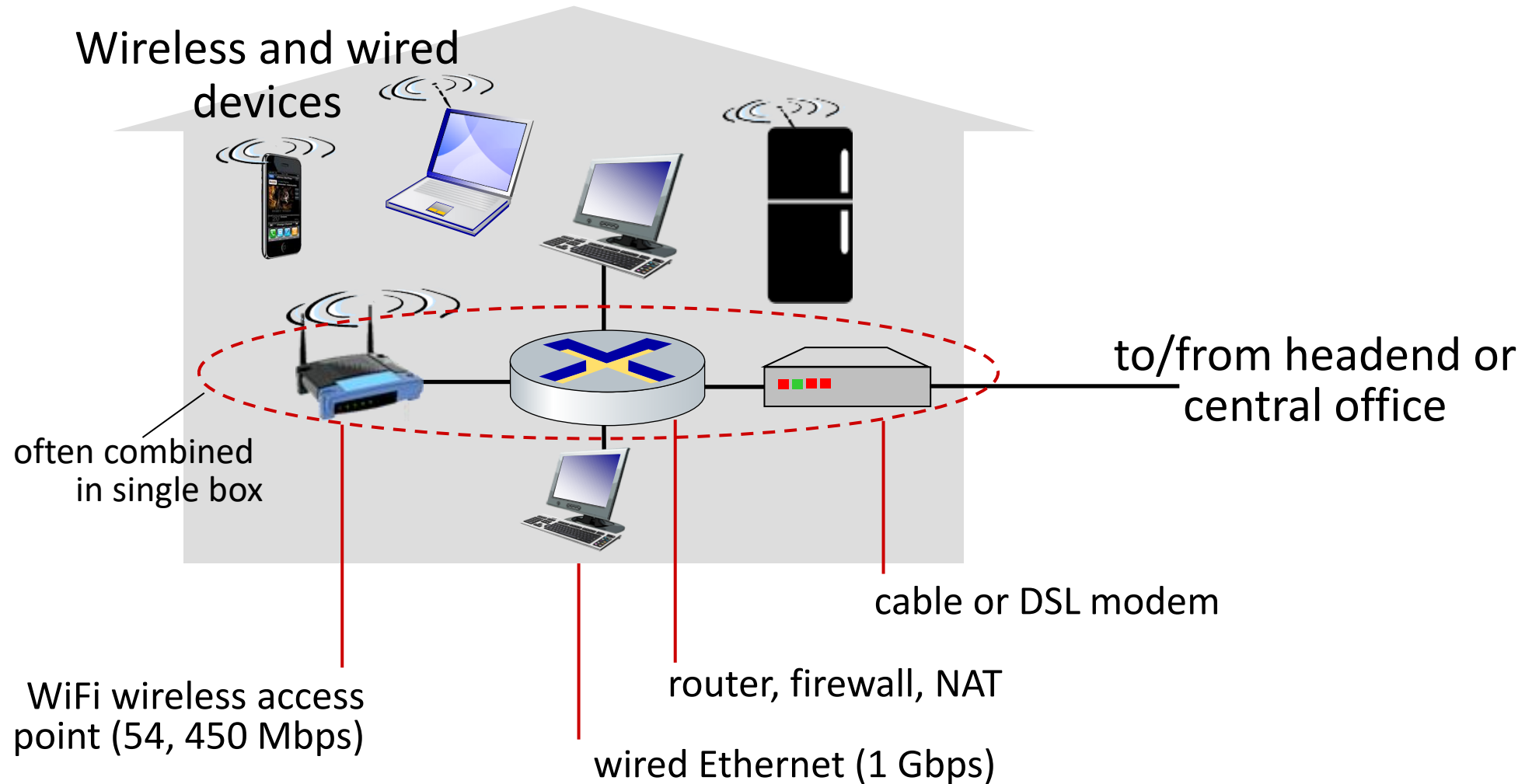
- homes *share access network* to cable headend

Access networks: digital subscriber line (DSL)



- use *existing* telephone line to central office DSLAM
 - data over DSL phone line goes to Internet
 - voice over DSL phone line goes to telephone net
- 24-52 Mbps dedicated downstream transmission rate
- 3.5-16 Mbps dedicated upstream transmission rate

Access networks: home networks



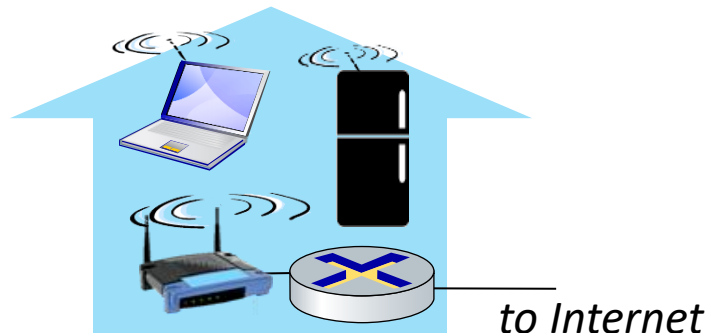
Wireless access networks

Shared *wireless* access network connects end system to router

- via base station aka “access point”

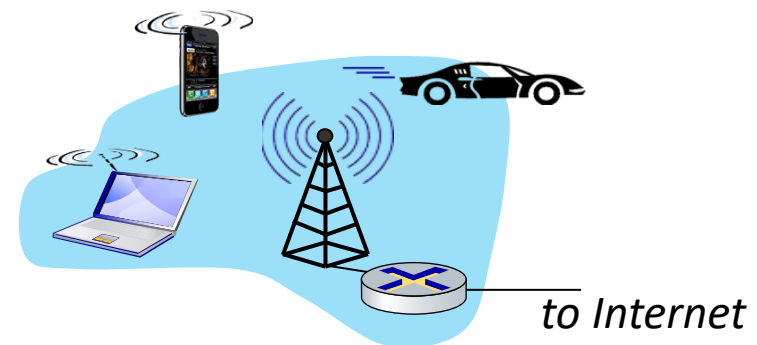
Wireless local area networks (WLANs)

- typically within or around building (~100 ft)
- 802.11b/g/n (WiFi): 11, 54, 450 Mbps transmission rate

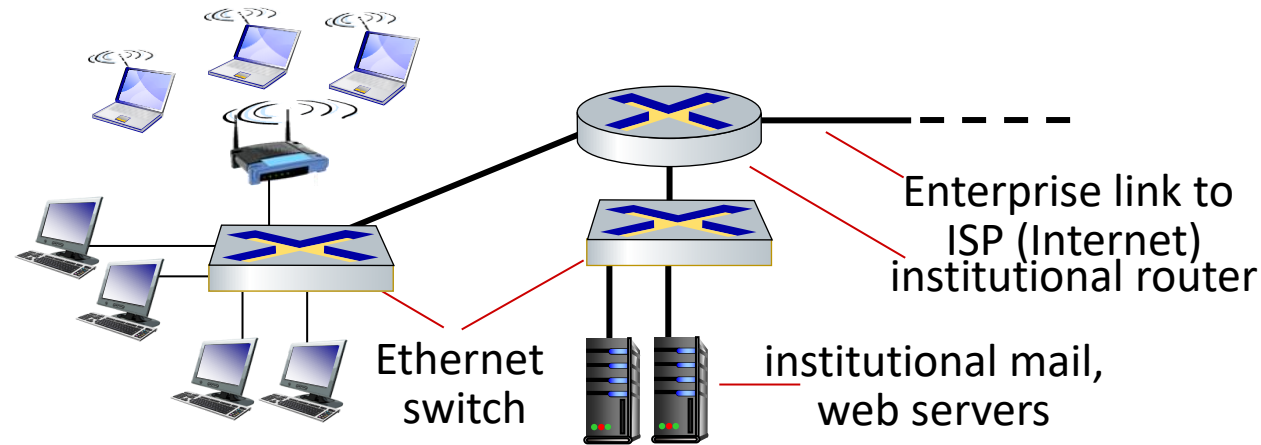


Wide-area cellular access networks

- provided by mobile, cellular network operator (10's km)
- 10's Mbps
- 4G/5G cellular networks



Access networks: enterprise networks



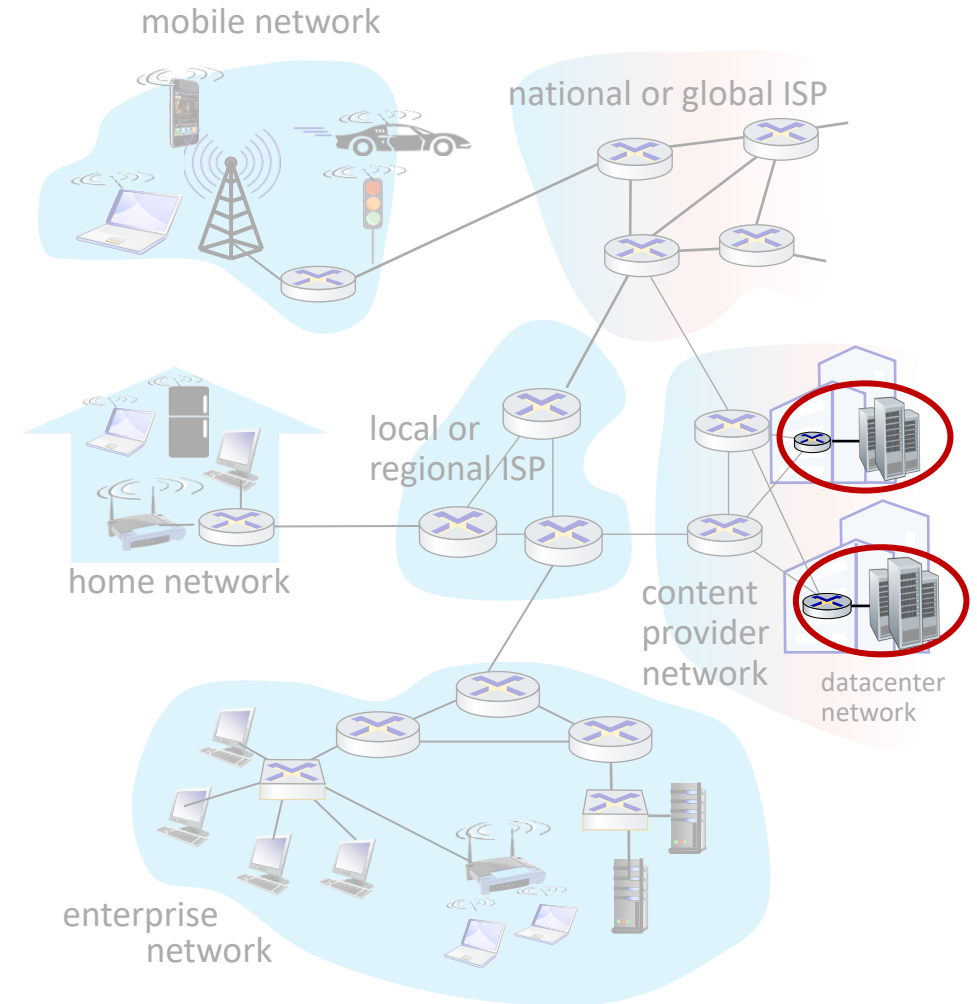
- companies, universities, etc.
- mix of wired, wireless link technologies, connecting a mix of switches and routers (we'll cover differences shortly)
 - Ethernet: wired access at 100Mbps, 1Gbps, 10Gbps
 - WiFi: wireless access points at 11, 54, 450 Mbps

Access networks: data center networks

- high-bandwidth links (10s to 100s Gbps) connect hundreds to thousands of servers together, and to Internet



Courtesy: Massachusetts Green High Performance Computing Center (mghpcc.org)



References

- Slides are modified version of slides provided by Kurose and Ross for their book